

Dataset: Bank Transactions (50 Records)

```
transactions = [
    "DEPOSIT",
    "WITHDRAW",
    "TRANSFER",
    "DEPOSIT",
    "BALANCE_CHECK",
    "TRANSFER",
    "DEPOSIT",
    "WITHDRAW",
    "LOAN_PAYMENT",
    "DEPOSIT",
    "TRANSFER",
    "WITHDRAW",
    "DEPOSIT",
    "BALANCE_CHECK",
    "TRANSFER",
    "DEPOSIT",
    "WITHDRAW",
    "LOAN_PAYMENT",
    "TRANSFER",
    "DEPOSIT",
    "WITHDRAW",
    "DEPOSIT",
    "TRANSFER",
    "BALANCE_CHECK",
    "WITHDRAW",
    "DEPOSIT",
    "TRANSFER",
    "LOAN_PAYMENT",
    "DEPOSIT",
    "WITHDRAW",
    "TRANSFER",
    "DEPOSIT",
    "BALANCE_CHECK",
    "TRANSFER",
    "WITHDRAW",
    "DEPOSIT",
    "TRANSFER",
```

```
"LOAN_PAYMENT",  
"DEPOSIT",  
"WITHDRAW",  
"TRANSFER",  
"BALANCE_CHECK",  
"DEPOSIT",  
"TRANSFER",  
"WITHDRAW",  
"LOAN_PAYMENT",  
"DEPOSIT",  
"TRANSFER",  
"BALANCE_CHECK",  
"DEPOSIT"  
]
```

```
rdd = spark.sparkContext.parallelize(transactions)
```

Basic RDD Exercises

1. Display all transactions.
2. Count total transactions.
3. Display first transaction.
4. Display first 15 transactions.
5. Display unique transaction types.
6. Count unique transaction types.
7. Sort transaction types alphabetically.
8. Convert all transaction names to lowercase.
9. Convert all transaction names to uppercase.
10. Display length of each transaction type.

Filter Exercises

11. Display only DEPOSIT transactions.
12. Display only WITHDRAW transactions.
13. Display only TRANSFER transactions.
14. Display only LOAN_PAYMENT transactions.
15. Display transactions containing CHECK.
16. Remove DEPOSIT transactions.
17. Remove TRANSFER transactions.
18. Display transactions longer than 8 characters.
19. Display transactions shorter than 10 characters.
20. Display transactions starting with B.

Map Exercises

21. Add prefix BANK_ to every transaction.
22. Add suffix _2025 to every transaction.
23. Convert transactions to tuple:
(transaction, length)
24. Convert transactions to tuple:
(transaction, 1)
25. Replace DEPOSIT with CASH_DEPOSIT.
26. Replace WITHDRAW with CASH_WITHDRAWAL.
27. Reverse each transaction name.

28. Extract first 3 characters from each transaction.

29. Extract last 4 characters from each transaction.

30. Create a transaction code:

DEPOSIT -> D

WITHDRAW -> W

TRANSFER -> T

Distinct and Frequency Analysis

31. Find distinct transaction types.

32. Count distinct transaction types.

33. Sort distinct transaction types.

34. Display distinct transaction types containing PAYMENT.

35. Display distinct transaction types containing CHECK.

36. Count occurrences of each transaction.

37. Find most frequent transaction.

38. Find least frequent transaction.

39. Display transaction frequencies.

40. Display transactions occurring more than 5 times.

Union and Intersection

```
extra_transactions = [  
    "DEPOSIT",  
    "ATM_WITHDRAWAL",  
    "TRANSFER",
```

```
"MOBILE_RECHARGE",  
"BALANCE_CHECK"  
]
```

```
rdd2 = spark.sparkContext.parallelize(extra_transactions)
```

41. Union rdd and rdd2.

42. Count records after union.

43. Display distinct records after union.

44. Find new transaction types introduced.

45. Count frequencies after union.

```
rdd3 = spark.sparkContext.parallelize([  
"DEPOSIT",  
"TRANSFER",  
"BALANCE_CHECK",  
"CREDIT_CARD_PAYMENT",  
"ATM_WITHDRAWAL"  
])
```

46. Find common transaction types.

47. Count common transaction types.

48. Sort common transaction types.

49. Display common transaction types containing CHECK.

50. Display common transaction types containing TRANSFER.

Numeric RDD Exercises

```
amounts = [  
500, 1200, 3000, 4500, 800, 950,  
1100, 2500, 4000, 6000,
```

```
750,1800,2200,3300,4700,  
5200,1500,2600,3700,4800,  
5900,6200,700,900,1400,  
1750,2100,2900,3600,4300,  
5100,5800,6500,7200,8100,  
9200,10500,12000,13500,15000,  
17000,19000,22000,25000,28000,  
30000,32000,35000,38000,40000  
]
```

```
amount_rdd = spark.sparkContext.parallelize(amounts)
```

51. Find total transaction amount.
52. Find average transaction amount.
53. Find maximum transaction amount.
54. Find minimum transaction amount.
55. Display amounts greater than 10000.
56. Display amounts less than 2000.
57. Add 500 to every amount.
58. Apply 10% bonus to every amount.
59. Sort amounts ascending.
60. Sort amounts descending.
61. Display top 10 highest amounts.
62. Display top 10 lowest amounts.
63. Count transactions above 5000.
64. Count transactions above 20000.
65. Find total of transactions above 10000.

FlatMap Exercises

```
customer_comments = [  
    "service was excellent",  
    "mobile banking is easy",  
    "transfer completed quickly",  
    "deposit process was smooth",  
    "excellent customer support",  
    "loan payment completed successfully",  
    "banking app is excellent",  
    "service is fast and reliable"  
]
```

```
comment_rdd = spark.sparkContext.parallelize(customer_comments)
```

66. Split all words using flatMap().
67. Count total words.
68. Display distinct words.
69. Sort words alphabetically.
70. Count frequency of each word.
71. Display words containing 'service'.
72. Display words containing 'excellent'.
73. Convert all words to uppercase.
74. Find longest word.
75. Find shortest word.
76. Find most frequently occurring word.
77. Find least frequently occurring word.
78. Count words starting with 'b'.

79. Count words ending with 'ly'.

80. Generate word frequency report.

Challenge Exercises

81. Find percentage of DEPOSIT transactions.

82. Find percentage of TRANSFER transactions.

83. Find percentage of WITHDRAW transactions.

84. Find transaction type with highest frequency.

85. Find transaction type with lowest frequency.

86. Create a transaction summary report.

87. Create a unique transaction report.

88. Create a frequency report.

89. Export unique transactions to text file.

90. Export transaction frequencies to text file.

91. Find total banking activities.

92. Find top 3 most common activities.

93. Find bottom 3 least common activities.

94. Display activities sorted by frequency.

95. Create a final banking analytics report.

96. Calculate total money processed.

97. Calculate average money processed.

98. Identify high-value transactions (> 20000).

99. Identify low-value transactions (< 1000).

100. Generate complete bank transaction dashboard output using RDD operat